DigiShow LINK

DigiShow is a lightweight cross-media control software designed for live performances and interactive art installations. It provides an intuitive console interface that enables signal control and mapping among various audio, lighting, robotic, and interactive devices.

Digital Artists	Create cross-media interactive art installations
Stage & Immersive Designers	Program cues to synchronize lighting, mechanical devices, and music
Musicians	Enhance live performances with DIY electronic instruments or automated lighting
Engineers & Makers	Develop smart interactive projects like home automations

Install and Run

macOS

Copy app "DigiShow" to your Applications folder and run it.

If you see the error message says **DigiShow app is damaged and can't be opened**, please also need to run this command in the terminal before starting the app for the first time.

```
xattr -cr /Applications/DigiShow.app
```

Windows

Copy folder "DigiShow LINK" to your disk and run "DigiShow.exe" in the folder.

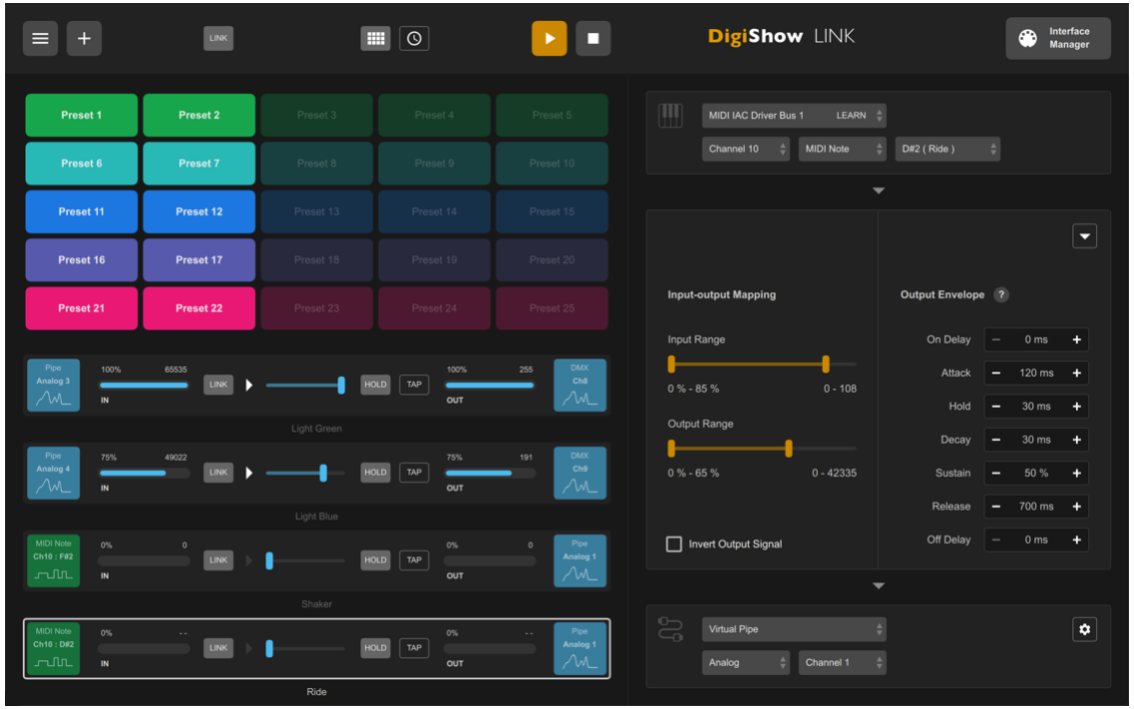
If you see the error message says **The code execution cannot proceed because MSVCP140.dll was not found**, please also need to run **Extra\vc_redist.x64.exe** to install Visual C++ Redistributable to your windows system.
It's also recommended to install **K-Lite Codec Pack** and **loopMIDI** to your Windows, the installer files can be found in Extra folder.

Key Features

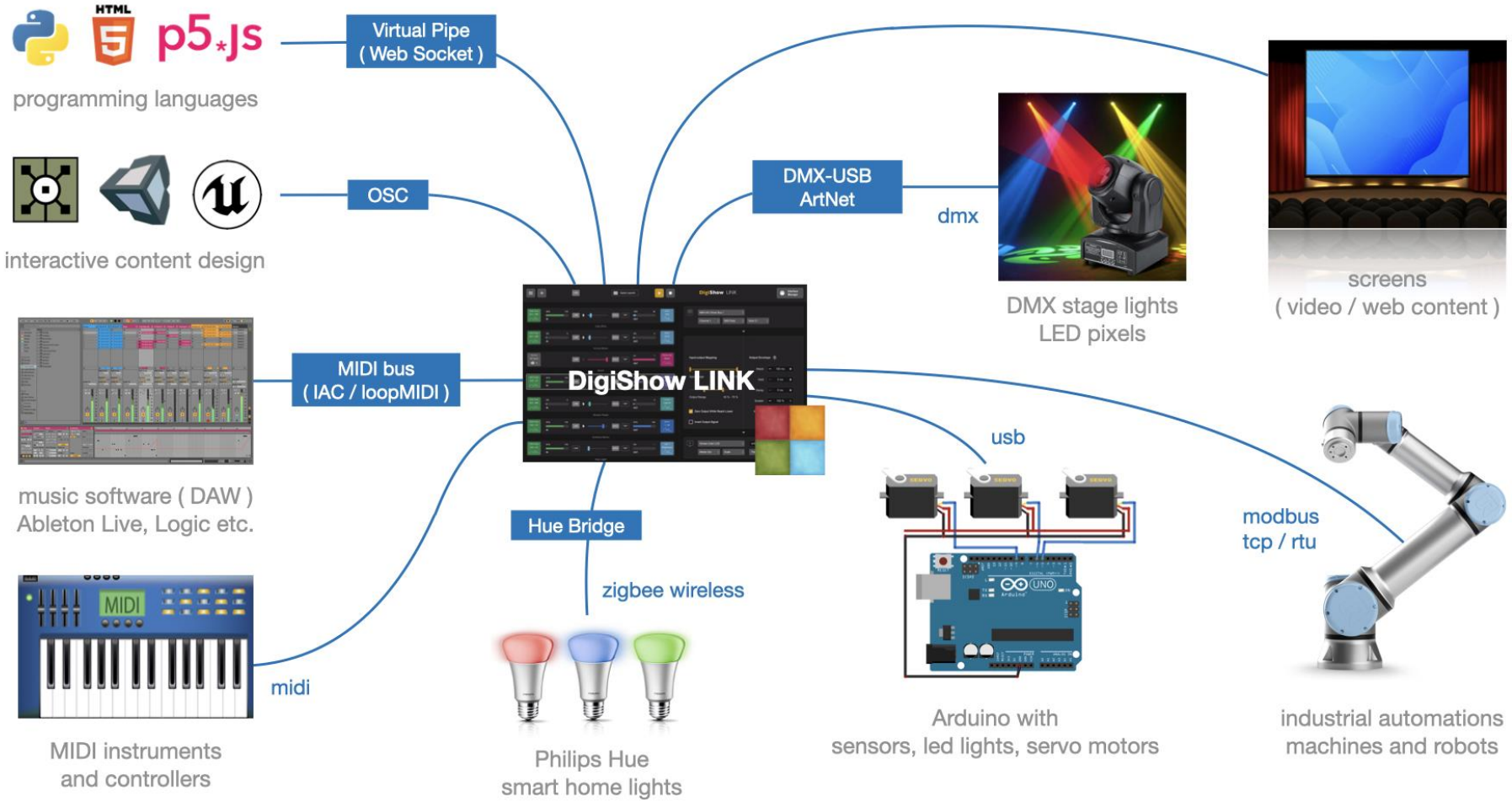
- **Multi-Protocol Support**
  Supports MIDI, DMX, OSC, Modbus, Arduino, Hue, and other protocols for coordinated control of audio, lighting, screens, robotics and more.

- **Signal Mapping**
  Converts MIDI notes, OSC control signals, etc., into lighting, servo motor, and media playback commands. Also transforms sensor inputs into MIDI/OSC signals for music software (e.g., Ableton Live, Logic Pro) and real-time visual creative tools (e.g., TouchDesigner, Unreal Engine).

- **Interactive Control** Ideal for DJ performances, stage or space lighting synchronization, experimental music, and interactive installations to enhance live engagement and visual effects.



DigiShow main window:
Preset launcher, signal link table and signal mapping settings.



For a typical 'digital' show, requires some particular digital things working together, along with DigiShow LINK.

Technologies and Functions

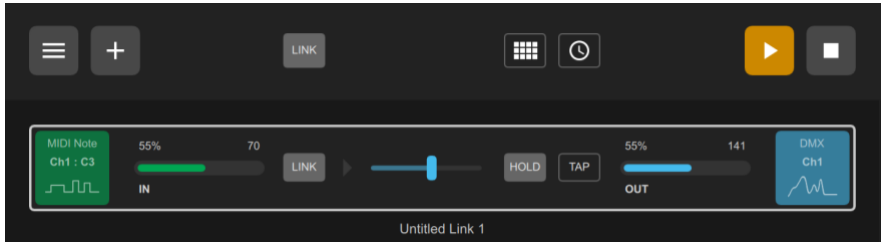
- **No-Coding Arduino**
  Provides a ready-to-use sketch drives mapping sensors and actuators connected to Arduino as inputs and outputs used directly in the DigiShow.

- **Cue Player**
  Features a graphical interface for designing output curves for specific scenarios, which can be attached to presets for synchronized playback.

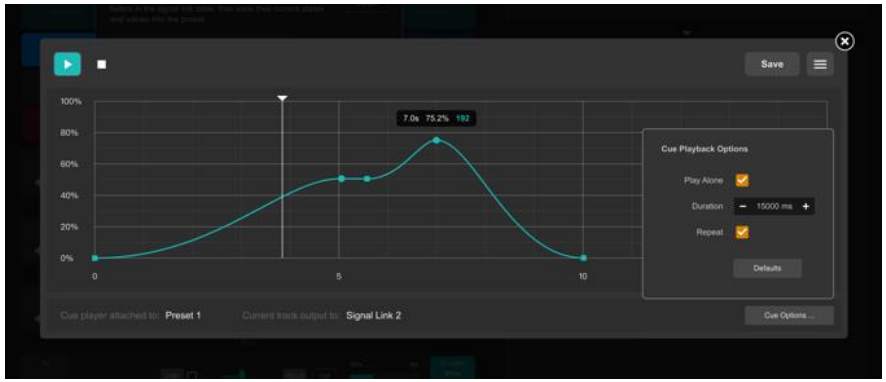
- **Pixel Mapping**
  Dynamically maps video or image pixels to lighting arrays for visual effects.

- **Scripting Support**
  Allows JavaScript/QML expressions and scripts for advanced interactive logic.

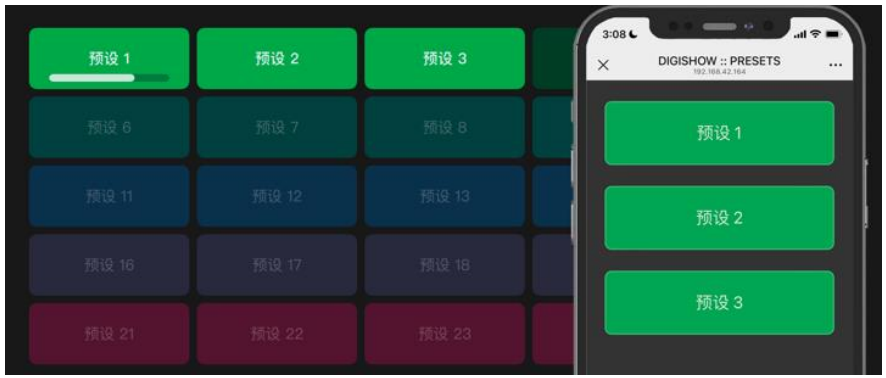
- **Network Protocols**
  Supports OSC and pipes over WebSocket for external extensions, as well as Art-Net for large-scale lighting control systems.



A signal bar linking a MIDI note to a DMX lighting channel.



Timeline editor for Cue Player: Design the output signal playback curve in the timeline.



Preset launch buttons in the DigiShow app and the mobile.

Examples

To better understand the features and capabilities of DigiShow, open the .dgs example projects in the examples folder using the installed DigiShow app:

1. **basic**

   Beginner-friendly examples covering keyboard hotkeys, virtual pipes, presets, cues, and signal mapping (e.g., MIDI mapping).

2. **audio player**

   Fun examples demonstrating how to make the most of the audio player interface.

3. **playback control**

   Understanding playback control is a great way to learn interactive system design. We provide a series of progressive examples.

4. **presets and beats**

   The Preset Launcher and Beat Maker are key DigiShow components. We demonstrate them with progressive lighting console examples.

5. **audio input**

   Learn how to connect a microphone or pickup via the Audio Input interface to obtain sound level (or vibration strength).

6. **dmx**

   Control lights using DMX and ArtNet, examples include 512-channel dimmer testing and pixel mapping.

7. **screen**

Dynamically display images or play videos within DigiShow's Screen interface.

8. **osc**

   Exchange signal variables with external software in real time via OSC. Examples include TouchDesigner, Unity3D, Blender, and Python (works with AI agent).

9. **web app**

   Build web apps that work with DigiShow, ideal for AI agent vibe coding.

10. **remote pipe**

    Explains how to use Remote Virtual Pipes to synchronize signals between two DigiShow instances, as well as communication with WebSocket clients in web pages or python scripts.

11. **javascript**

    Use JavaScript expressions and scripts in DigiShow to implement more intelligent logic.

12. **ableton**

    A complete demo using DigiShow with Ableton Live for making music installations.

13. **wireless**

    Build interactive wireless gadgets for DigiShow using ESP32 Arduino (via WiFi and Bluetooth).

For more learning, please read DigiShow Tutorials
https://github.com/robinz-labs/digishow/blob/master/guides/tutorials.md

Release Downloads

Please visit https://github.com/robinz-labs/digishow/releases/latest to download the latest releases:

- DigiShow LINK for windows (64bit Intel)	digishow_win_x.x.x_x64.zip
- DigiShow LINK for macOS (64bit Intel)	digishow_mac_x.x.x_x64.zip
- DigiShow LINK for macOS (64bit Apple M series)	digishow_mac_x.x.x_arm64.zip
- DigiShow LINK for Linux (Raspberry Pi or 64bit ARM)	digishow_linux_x.x.x_arm64.zip

Go to the page, where choose the file in Assets list to download.

Extra Downloads and Resources

Arduino DigiShow RIOC library

Required to enable DigiShow LINK app to control sensors and actuators connected on your Arduino.
Find and install DigiShow RIOC in the library manager of Arduino IDE, or download it from github.
https://github.com/robinz-labs/rioc-arduino/releases

MIDI Virtual MIDI bus drivers (IAC / loopMIDI)

In order to communicate with MIDI messages between DigiShow LINK and other software, users just need to setup a virtual MIDI bus in the operation system.

IAC for Mac
https://help.ableton.com/hc/en-us/articles/209774225-How-to-setup-a-virtual-MIDI-bus

loopMIDI for Windows
http://www.tobias-erichsen.de/software/loopmidi.html

DMX ENTTEC DMX USB Pro driver (FTDI VCP)
Required to enable DigiShow LINK to control DMX lightings through an ENTTEC adapter.
https://www.ftdichip.com/Drivers/VCP.htm

Screen K-Lite Codec Pack (for windows)
Required to enable DigiShow LINK to play MP4, MOV video files on your Windows computer.
https://www.codecguide.com/download_kl.htm

Developer Resources

DigiShow is open-source.
If you would like to rebuild this software using the source code we contributed, please visit https://github.com/robinz-labs/digishow .
Building executables from source using the qmake tool or the QtCreator application requires Qt 5.12 or 5.15 LTS.

Additional library dependencies are already included in the repository:

- RtMidi 4.0.0 http://www.music.mcgill.ca/~gary/rtmidi/
- TinyOSC library https://github.com/mhroth/tinyosc
- Ableton Link library https://ableton.github.io/link/
- global hotkey library https://github.com/Skycoder42/QHotkey

The source code can be compiled for target platforms compatible with:

- macOS 10.13 or higher
- windows 10 or windows 11 (64-bit version)
- linux (see the websites of Qt and other dependent libraries for compatibility details)